

Student ID 1 & 2: (Do this in pairs)

A Proposed Reduction from 3SAT to Vertex Cover.

Given a 3-CNF formula ϕ with variables $x_1, \dots, x_n$ and clauses $C_1, \dots, C_m$, construct a graph $G = (V, E)$ and integer k as follows:

1. **Variable gadgets.** For each variable x_i , add two nodes labeled x_i and $\bar{x}_i$ connected by an edge.
2. **Clause gadgets.** For each clause $C_j = (\ell_{j,1} \vee \ell_{j,2} \vee \ell_{j,3})$, add a single node c_j and add three edges connecting c_j to the literal nodes $\ell_{j,1}$, $\ell_{j,2}$, and $\ell_{j,3}$ already in the graph (from the variable gadgets).
3. Set $k = n + m$.

Claimed correctness: ϕ is satisfiable if and only if G has a vertex cover of size $\leq k$.

1. Construct the VC instance.

Consider the following 3-CNF formula over variables x_1, x_2, x_3 :

$$\phi = (x_1 \vee x_2 \vee x_3) \wedge (\bar{x}_1 \vee x_2 \vee x_3) \wedge (x_1 \vee \bar{x}_2 \vee x_3) \wedge (\bar{x}_1 \vee \bar{x}_2 \vee \bar{x}_3).$$

Apply the proposed reduction to ϕ and draw the resulting graph G . List all nodes and edges explicitly, and state the value of k .

2. Verify the YES instances.

- (a) Show that ϕ is a YES instance of 3SAT by exhibiting a satisfying truth assignment and verifying each clause.
- (b) Using the vertex cover S induced by the satisfying assignment from part (a) together with all clause nodes, verify that S is a valid vertex cover of size k in the graph G from Problem 1.

3. Prove the reduction is incorrect.

The reduction above contains a logical error. Identify and explain the error.